

Safal Kumar Shrestha

AI/ML Engineer | Computer Vision, AI Agents and Software Development

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SUMMARY

AI undergraduate and developer combining full-time study with part-time product development. Contributed to 3 shipped applications, including KU Krishi (180+ users) and TrekSathi (150+ users). Secured STEAM research funding and DFRobot sponsorship for Hydroponics AI, building 6 sensing/control workflows across 2 ESP32 platforms. Experience spans computer vision, AI agents, full-stack applications, and self-hosted infrastructure.

EDUCATION

Bachelor of Technology (B.Tech) in Artificial Intelligence (Full-time)

Kathmandu University, School of Engineering, Nepal | 2022 - Present

Expected graduation: 2027 | **CGPA: 3.81**

Coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Large Language Models, Probability and Statistics, Data Structures and Algorithms.

Diploma in Computer Engineering

Nepal Polytechnic Institute, Nepal | Completed: 2022

Final percentage: 81.62%. Coursework: C/C++, Java, Computer Architecture, Operating Systems, Digital Logic.

TECHNICAL SKILLS

Programming: Python, SQL, Java, Kotlin, C/C++

Machine Learning: PyTorch, TensorFlow, scikit-learn, deep learning, neural networks, data preprocessing, model training, evaluation, error analysis, reinforcement learning

Computer Vision: OpenCV, YOLOv8, Tiny U-Net, image processing, object detection, segmentation

Generative AI: Large language models (LLMs), LangGraph, AI agents, tool calling, prompt engineering; retrieval-augmented generation (RAG), embeddings, and vector database concepts

Web, Mobile and Design: Next.js, Node.js, Flutter, Figma, UI/UX

Backend and Data: FastAPI, Django REST Framework, REST APIs, webhooks, MySQL, MongoDB, Firebase Firestore

DevOps and Systems: Docker, Kubernetes, Linux, Git/GitHub, HAProxy, model serving, testing, debugging, deployment, monitoring, logging, SSL/TLS, reverse proxies, firewalls, technical documentation

IoT and Simulation: ESP32, MQTT, WebSockets, sensor integration, actuator control; CARLA exploration

WORK EXPERIENCE

AI/ML and Product Developer (Part-time)

bala.ai | Remote | 2022 - Present

- Contributed Python services, data preparation, model integration, testing, releases, and monitoring for 3 shipped apps: KU Krishi, TrekSathi, and Tiffin, alongside full-time study.
- Develop computer-vision and AI-agent workflows; investigate failures using test inputs, logs, and user feedback.
- Operate 5 self-hosted workloads supporting AI services, application backends, analytics, storage, and synchronization.
- Integrate REST APIs and SQL/NoSQL databases; secure endpoints using SSL/TLS, reverse proxies, and firewalls.

Machine Learning, Computer Vision, and IoT Researcher (Part-time)

Hydroponics AI, STEAM Research Grant, Kathmandu University, Nepal | 2026

- Secured STEAM research funding and DFRobot hardware sponsorship for a reinforcement-learning-driven hydroponics automation prototype.
- Built 6 sensing/control workflows across 2 ESP32 platforms for environmental monitoring, pumping, and nutrient dosing; integrated pH, electrical conductivity, and carbon dioxide readings via MQTT.
- Developed FastAPI, MQTT, and WebSocket interfaces connecting sensor ingestion, inference, actuator control, and live monitoring; tested communication and control behavior and documented observations.

High-Performance Computing Core Engineer (Part-time)

AI HPC Initiative, Kathmandu University, Nepal | 2025

- Configured Kubernetes and HAProxy to route workloads and support distributed AI workers.
- Profiled NVIDIA RTX 3080 Ti topology, allocation behavior, and virtualization limits; documented 3 constraint categories—hardware, team knowledge, and physical infrastructure—to guide deployment decisions.

Software Engineer, Android (Full-time)

Shop2App Corp | United States, Remote | Feb 2022 - Jul 2022

- Built Java/Kotlin commerce features and integrated Shopify APIs and webhooks to synchronize merchant, mobile, and backend data.
- Collaborated in a distributed team on testing, debugging, deployment, and Google Play releases.

PROJECTS

TrekSathi: Tourism Application and AI Agent | Jul 2025 - Present

- Built a LangGraph tool-calling agent combining user input and trail context with geospatial and weather services to support route planning.
- Integrated Python, Django REST Framework, and MongoDB workflows for trails, waypoints, elevation, weather, and offline maps in an application used by more than 150 people.

KU Krishi: Computer Vision Agricultural Assistant

- Built a 4-stage workflow using TensorFlow, YOLOv8, and OpenCV: image preprocessing, crop-disease inference, FastAPI model serving, and Flutter delivery.
- Published on Google Play and reached 180+ users; selected as a semi-finalist for Nepal's 2024 National ICT Award.

Satellite-Image Land Segmentation | Jun 2026 - Jul 2026

- Preprocessed satellite imagery with Python and compared segmentation outputs for geographic information system (GIS) analysis and model experimentation.

Wilms Tumor Segmentation | Jun 2025 - Jul 2025

- Developed a Tiny U-Net neural network to delineate tumor regions in medical images for a Pattern Recognition course project.

PixShop: Multi-Store E-commerce Website | Sep 2023 - May 2024

- Built a full-stack website with a shared Next.js interface and Node.js backend services supporting multiple shops and their commerce workflows.

Tiffin: Food Delivery Platform | Sep 2023 - Feb 2024

- Developed a Flutter app for chefs and clients across 4 workflows: discovery, pre-orders, restaurant browsing, and delivery.
- Connected Firebase Firestore order streams, Google Maps radius matching, and a Node.js backend for user management, order history, analytics, and mobile APIs.

ONGOING RESEARCH

Autonomous Driving with CARLA | Ongoing exploration

- Explore reinforcement learning for autonomous vehicle decision-making and control, emphasizing safe, efficient, and traffic-compliant behavior in simulation.

Reinforcement Learning for Software-Defined Networking | Ongoing exploration

- Explore reinforcement learning for Software-Defined Networking (SDN), focusing on intelligent network control, optimization, and decision-making.

LEADERSHIP

Operations Lead

AI Convergence, Kathmandu University, Nepal | 2025 - Present

- Coordinate 4 operational areas for university AI events: teams, logistics, schedules, and technical requirements; communicate priorities, progress, blockers, and handoffs.
- Coordinated teams and logistics for National AI Summit (Aug 2025) and AI Conclave (Jan 2026).

CERTIFICATIONS

LangGraph Certificate - LangChain Academy.

Python Instructor Certificate - KUIC.

AWARDS

STEAM Research Grant and DFRobot Hardware Sponsorship - Hydroponics AI, 2026.

National ICT Award Semi-Finalist - KU Krishi, 2024.

AI Crusade, Third Place - 2024.